Research Topic

Exploring the Effect of Agriculture GDP(%) on Education Disparities (Financial, Gender and Region) Across Education Levels (Primary, Lower Secondary and Upper Secondary) in Countries with Major Employment(%) in Agriculture from 2005 till 2023

Research Introduction and Background

All countries face education disparities between various categories, such as socio-economic status, gender, region etc. These disparities have been extensively researched in isolation, as well as in connection to other factors, such as economic growth (Hanushek & Wößmann, 2007). However, I found no research on how agricultural advancement affects educational disparities. This research topic is inspired by the following concepts, proven by literature:

- Educational disparities manifest most visibly between socio-economic statuses (Pokropek, et al., 2015).
- The livelihood of most people belonging to lower socio-economic statuses is dependent on agriculture in some way or another (SHORT, 2002)

Therefore, one may presume that the liveloods of poor people improve with a higher GDP percentage from agriculture. This, in turn, may reduce educational disparities between socioeconomic statuses, as well as other categories (region, gender etc.) However, this may also lead to a reverse increase in educational disparities, as children may forego education to provide agricultural labour. Alternatively, agricultural GDP(%) may have absolutely no effect on educational disparities.

The effect of agricultural GDP on education disparities may manifest visibly in countries where a large percentage of employment is in agriculture. Hence, for the purpose of this analysis, countries with agricultural employment greater than 35% are considered. For this analysis, years from 2005 till 2023 will be considered.

This research topic is extremely unique. Although agricultural GDP and education disparities have been extensively researched in isolation, I did not find a single research paper linking the two. This is surprising because the two concepts that inspired this topic lead to the idea that agricultural GDP(%) must have some effect on education disparities, at least in countries with a high employment percentage in agriculture.

Apart from being unique, this topic is relevant to the ongoing global discussion about how to reduce education disparities. If this analysis finds a connection between agriculture GDP(%) and education disparities, it could contribute to government policy.

Aim

Exploring the relationship between agriculture GDP percentage and education disparities (financial, gender and region) across education levels (primary, lower secondary and upper secondary) in countries with a major employment percentage in agriculture from 2005 till 2023

Objectives

- Acquire data for agricultural GDP(%), agricultural employment(%) and educational outcomes for various countries
- Perform exploratory data analysis on the acquired data
- Shortlist countries with major employment(%) in agriculture
- For the shortlisted countries, analyze the relationship between disparities across education levels

- For the shortlisted countries, analyze the relationship between agricultural GDP(%) and disparities across education levels

Scope

- This analysis uses completion rate as a metric to measure education outcomes. It does not incorporate other metrics such as attendance rate,enrollment rate, and dropout rate because completion is a closer measure of educational outcomes. However, for a project of a larger scope, each of the metrics could be explored for a more thorough analysis
- This analysis focuses on three education levels: primary, lower secondary and upper secondary. It does not incorporate levels after these, such as higher education, because higher levels have much lesser education outcomes. Furthermore, the effect of agricultural GDP or other factors on higher education levels would be less apparent. Therefore, these 3 levels are chosen for a clear analysis. However, if there was no time or size limition, further levels could be explored in depth to draw more interesting insights
- This analysis focuses on education disparities between three groups: financial, region (urban/rural) and gender. Education disparities generally show up the most between financial statuses. Following this, education disparities can also be detected between urban/rural regions and genders. This analysis does not explore other kinds of disparities e.g. cultural, religious because they are less apparent and difficult to quantify.
- This analysis focuses on a timeframe from 2005 till 2023. This timeframe is recent enough to make the analysis relevant, but it is also broad enough to draw interesting insights.
- This analysis averages the values from all years. It does not perform a time series analysis for each year, due to the limited scope of this project. However, a time series analysis could draw really interesting trends between yearly agricultural GDP and educational disparities.

Data Processing Pipeline

1. **Processing Agriculture Data**
   First, the agriculture dataset must be cleaned and filtered to contain only those countries which have a high employment percentage in agriculture
   A. **Loading Data**
      First, the csv dataset must be loaded into a pandas dataframe
   B. **Assessing Data**
      The structure of the dataset should be assessed
   C. **Cleaning Data**
      Initial cleaning steps should be carried out, such as removing unneccessary rows/columns, removing duplicates, removing outliers, ensuring that the data is in the expected form etc.
   D. **Structuring Data for Purpose**
      After cleaning the dataset, it should be structured to facilitate the purpose of the analysis. This step includes filtering the dataframe so it contains only those countries which have a high employment percentage in agriculture.
2. **Processing Education Data**
   After the agriculture dataset has been cleaned and structured, the education dataset should be processed
   A. **Loading Data**
      First, the csv dataset must be loaded into a pandas dataframe
   B. **Assessing Data**
      The structure of the dataset should be assessed
   C. **Cleaning Data**
      Initial cleaning steps should be carried out, such as removing unneccessary rows/columns, removing duplicates, removing outliers, ensuring that the data is in the expected form etc.
   D. **Modifying and Structuring Data for Purpose**
      After cleaning the dataset, it should be structured in a way to facilitate visualizations and analysis. For example, it could be seperated into different dataframes for different education levels/disparities.
3. **Visualizing and Analyzing Data**

After both datasets have been cleaned and structured, the data can be visualized with bar graphs or scatter plots. At each education level, disparities can be visualized with respect to each other or with respect to the agricultural GDP(%). This will draw interesting insights into the research topic.

4. **Evaluating Results**
   The results of the exploratory data analysis will be critically evaluated.

Data Acquisition

Datasets: Sources, Relevance and Format

1. **Agriculture Dataset**
   This dataset was downloaded from the World Bank database 'World Development Indicators' (World Bank, n.d.). This database is compiled by the World Bank from trusted international sources, and it is consistently updated. It has a CC BY 4.0 license.

   The database provided the following options for downloading a dataset:
   - **Country:** I selected all countries. The countries will be filtered during the exploratory analysis to include only the countries with major agricultural employment percentages
   - **Series:** I selected two indicators:
     - **Agriculture, forestry, and fishing, value added (% of GDP)**
       This indicator is useful as it provides a measure of the agricultural GDP(%). This analysis explores the effect of agricultural GDP(%) on educational disparities.
     - **Employment in agriculture (% of total employment) (modeled ILO estimate)**
       This indicator is useful as the analysis will focus on countries with major employment percentage in agriculture. This indicator will help filter out those countries
   - **Time:** I selected all years from 2005 till 2023. This range is recent enough to make the analysis relevent, but it is also broad enough to account for a wide range of data.

2. **Education Dataset**
   This dataset was downloaded from the World Bank database 'Education Statistics - All Indicators' (World Bank, n.d.). This database is compiled by the World Bank from trusted international sources, and it is consistently updated. It has a CC BY 4.0 license.

   The database provided the following options for downloading a dataset:
   - **Country:** I selected all countries. The countries will be filtered during the exploratory analysis to include only the countries with major agricultural employment percentages
   - **Series:** I selected 60 series (1 x metric x 3 x education levels x 2 x regions x 5 x financial quintiles x 2 x genders). The series strings contain comma-seperated categories. I chose combinations of the following categories.
     A. **Education metric**
        a. Completion rate
           This metric is useful for analyzing disparities in education outcomes. When downloading the dataset, the completion rate is chosen over other metrics such as attendance rate/enrollment rate because completion is a closer measure of educational outcomes.
     B. **Education level**
        a. Primary
        b. Lower secondary
        c. Upper secondary
        This category is useful for analyzing disparities at primary, lower secondary and upper secondary education levels. Levels after these, such as higher education, have much lesser education outcomes and the effect of agricultural GDP would also be less apparent. Therefore, these 3 levels are chosen for a clear analysis.
     C. **Region**
        a. Urban

    b. Rural

   This category is useful for analyzing urban/rural disparities

   D. **Financial status (quintile)**
      a. Poorest quintile
      b. Second quintile
      c. Middle quintile
      d. Fourth quintile
      e. Richest quintile

   This category is useful for analyzing financial disparities.

   E. **Gender**
      a. Male
      b. Female

   This category is useful for analyzing gender disparities

- **Time:** I selected all years from 2005 till 2023. This range is recent enough to make the analysis relevent, but it is also broad enough to account for a wide range of data.

Both datasets were downloaded from the World Bank in CSV format. These files are converted to dataframes using Pandas, and numerical analysis is performed on them. Both datasets are used in correspondence, for example the country names in both datasets will be used when filtering out countries with a high employment percentage in agriculture. Therefore, downloading both datasets from the World Bank ensures consistency in handling similar data.

Alternative Dataset Sources

Education Dataset	Agriculture Dataset
UNESCO Institute of Statistics 1. **Strengths** The World Bank compiles education data from the UNESCO Institute of Statistics. Therefore, directly accessing the education dataset from there may be the most updated version. 2. **Weaknesses** The Unesco Institute of Statistics does not have corresponding datasets for agriculture GDP and employment. Hence, if I took the education dataset from there, it might show inconsistencies with agricultural data taken from somewhere else	International Labour Organization 1. **Strengths** The International Labour Organization contains data for employment in agriculture. The World Bank compiles employment data from the ILO. Therefore, directly accessing this data from the ILO would give the most updated version 2. **Weaknesses** The ILO does not have a corresponding dataset for agricultural GDP
National Center for Education Statistics 1. **Strengths** The National Center for Education Statistics contains education data for localized regions. Using this data would enable me to analyze regional educational disparities in more detail. 2. **Weaknesses** The National Center for Education Statistics also does not have corresponding datasets for agriculture GDP and employment. Hence, if I took the education dataset from there, it might show inconsistencies with agricultural data taken from somewhere else	United Nations Food and Agriculture Organization 1. **Strengths** The UN Food and Agriculture Organization contains datasets for employment in agriculture as well as agricultural GDP. This data is updated regularly 2. **Weaknesses** This data does not have corresponding education datasets. Hence, if I took the agriculture dataset from there, it might show inconsistencies with education data taken from somewhere else

Data Exploration

Importing libraries required for exploration

In [1]:
```
# this library is used to create and analyze dataframes
import pandas
# this library is be used to include HTML in print statements
from IPython.display import HTML
# this library is used for plotting graphs
import matplotlib.pyplot as plt
# this library is used for calculations like
# determining the equations of lines of best fit
# on scatter plots
import numpy as np
```

Processing Agriculture Dataset

Loading Agriculture Dataset

In [2]:
```
# creating a dataframe from the csv file
df_agriculture = pandas.read_csv("archive/agriculture_data.csv")

pandas.set_option('display.max_colwidth', None)

#assessing the structure of the dataframe
df_agriculture
```

Out[2]:

	Country Name	Country Code	Series Name	Series Code	2005 [YR2005]	2006 [YR2006]	2007 [YR2007]	2
0	Afghanistan	AFG	Employment in agriculture (% of total employment) (modeled ILO estimate)	SL.AGR.EMPL.ZS	61.6371750441572	60.8719056790228	59.8526540961945	59.0
1	Afghanistan	AFG	Agriculture, forestry, and fishing, value added (% of GDP)	NV.AGR.TOTL.ZS	31.114854912062	28.6359685844686	30.1050113574013	24.8
2	Africa Eastern and Southern	AFE	Employment in agriculture (% of total employment) (modeled ILO estimate)	SL.AGR.EMPL.ZS	63.793870084647	63.1336708563811	62.5749968146997	62.2
3	Africa Eastern and Southern	AFE	Agriculture, forestry, and fishing, value added (% of GDP)	NV.AGR.TOTL.ZS	9.76088731295034	10.0005620614351	10.9450277797119	12.1
4	Africa Western and Central	AFW	Employment in agriculture (% of total employment) (modeled ILO estimate)	SL.AGR.EMPL.ZS	54.4214969829428	53.8577991327742	53.0314046193462	52.2
...	...	...	...	...	...	...	...	
532	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
533	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
534	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
535	Data from database: World Development Indicators	NaN	NaN	NaN	NaN	NaN	NaN	
536	Last Updated: 12/16/2024	NaN	NaN	NaN	NaN	NaN	NaN	

537 rows × 23 columns

Cleaning Agriculture Dataset

The following observations can be made from the dataframe displayed above

Structure

Rows	Columns
For each country, there is a row for each of the following indicators: 1. **Employment in agriculture (% of total employment) (modeled ILO estimate)**	The following columns can be observed in the dataframe displayed above 1. **Country Name** 2. **Country Code**

Rows	Columns
2. Agriculture, forestry, and fishing, value added (% of GDP)	**3. Series Name** This is one of the following indicators: Employment(%) in agriculture GDP(%) of agriculture, forestry or fishing **4. Series Code** **5. Years from 2005 till 2023**

Initial cleaning steps

After viewing the dataframe, I outlined the following steps to clean and validate the data

1. The year columns represent numerical values. However, some values are not numeric. For example, the string '..' is used to represent empty values. Therefore, all values under the year columns must be coerced to numerical values to ensure consistency
2. Series codes and country codes are unnecessary to this analysis. Therefore, these columns must be deleted for simplicity
3. There are unnecessary rows towards the end of the dataframe, which contain metadata or NaN values. These rows must be deleted
4. If an entire row for an indicator contains no numerical values, it must be deleted as it is unusable
5. For each country, there must be a row for each of the following indicators
   A. Employment in agriculture as a percentage of the total employment
   B. GDP(%) from agriculture, forestry and fishing
   For this analysis, both indicators are used in correspondance. For a certain country, if there is no data for one indicator, the data on the other indicator is also unusable. Hence, countries which do not have two rows in the dataframe should be removed.
6. If a row has the same country name and series name (indicator), that row is a duplicate and should be removed

In [3]:
```
#creating a seperate dataframe so the steps in the data processing pipeline can be tracked back
df_agriculture_numeric = df_agriculture.copy(deep = True)

#coercing values under the year columns (5th column onwards) to numeric
df_agriculture_numeric.iloc[:, 4:] = df_agriculture.iloc[:, 4:].apply(
    pandas.to_numeric,
    errors='coerce'
)

#deleting the unnecessary 'Series Code' and 'Country Code' columns
df_agriculture_codes_removed = df_agriculture_numeric.drop (
    ['Series Code', 'Country Code'], # series code and country code columns
    axis=1, # column axis
)

# removing unnecessary rows that have only NaN values under the year columns
# this removes the unwanted rows towards the end of the dataframe,
# as well as any indicator rows that have no numerical values
df_agriculture_empty_rows_removed = df_agriculture_codes_removed.dropna (
    subset = df_agriculture_codes_removed.columns[2:], # year columns
    how='all' # row must be removed if all the year columns values are NaN
)

# removing any duplicate rows (with the same country name and series name)
df_agriculture_duplicates_removed = df_agriculture_empty_rows_removed.drop_duplicates (
    subset=['Country Name', 'Series Name'] # rows with the same country name and series name
)

# filtering the dataframe so that there are two rows (one for each indicator) representing each country
df_agriculture_two_indicators =df_agriculture_duplicates_removed.groupby('Country Name').filter(
    lambda c: len(c) == 2
)

#after dropping rows, resetting the index values so they appear from 0 in sequence
df_agriculture_two_indicators.reset_index(drop=True, inplace=True)
```

```
#displaying the dataframe again to view the result
df_agriculture_two_indicators
```

Out[3]:

	Country Name	Series Name	2005 [YR2005]	2006 [YR2006]	2007 [YR2007]	2008 [YR2008]	2009 [YR2009]	2010 [YR2010]	2011 [YR2011]	2012 [YR2012]
0	Afghanistan	Employment in agriculture (% of total employment) (modeled ILO estimate)	61.637175	60.871906	59.852654	59.00254	56.722689	54.34942	52.79091	50.416688
1	Afghanistan	Agriculture, forestry, and fishing, value added (% of GDP)	31.114855	28.635969	30.105011	24.89227	29.297501	26.210069	23.743664	24.390874
2	Africa Eastern and Southern	Employment in agriculture (% of total employment) (modeled ILO estimate)	63.79387	63.133671	62.574997	62.220835	61.889076	61.2885	60.337927	59.540712
3	Africa Eastern and Southern	Agriculture, forestry, and fishing, value added (% of GDP)	9.760887	10.000562	10.945028	12.191482	12.507847	10.906134	10.113029	10.83928
4	Africa Western and Central	Employment in agriculture (% of total employment) (modeled ILO estimate)	54.421497	53.857799	53.031405	52.2095	51.268941	50.308216	49.742822	48.864786
...	...	...	...	...	...	...	...	...	...	...
455	Yemen, Rep.	Agriculture, forestry, and fishing, value added (% of GDP)	11.093242	10.705582	10.419289	10.196859	10.01907	9.873709	9.752644	16.990332
456	Zambia	Employment in agriculture (% of total employment) (modeled ILO estimate)	70.946222	70.418555	69.762439	69.008031	67.734663	66.488724	65.462888	63.933902
457	Zambia	Agriculture, forestry, and fishing, value added (% of GDP)	14.588551	13.206271	12.105596	11.454	11.552784	9.420946	9.648058	9.32164
458	Zimbabwe	Employment in agriculture (% of total employment) (modeled ILO estimate)	72.885354	72.991665	72.915384	72.224295	69.822497	67.162289	65.862729	65.902062
459	Zimbabwe	Agriculture, forestry, and fishing, value added (% of GDP)	17.14824	19.230116	21.197689	19.021074	10.74255	9.609863	8.665865	8.044518

460 rows × 21 columns

Replacing Outliers and Storing Mean Values

The following steps are taken to replace outliers and calculate the mean value

- As the values are percentages, they must lie between 0 and 100. If a value under the year columns does not lie within this range, it ise replaced with NaN
- Two new columns are created to store the mean and standard deviation
- A nested loop iterates over all the values in the year columns. If any value lies more than 3 standard deviations away from the mean, it is replaced
- After the outliers have been replaced, the standard deviation column is removed and the mean column is recalculated
- The year columns are removed after calculating the mean, as they are no longer required

In [4]:

```
# creating a new dataframe to store agricultural data with outliers replaced,
# so that the previous steps in the pipeline are preserved
df_agriculture_outliers_removed = df_agriculture_two_indicators.copy(deep = True)

# if a value under the year columns is lesser than 0 or greater than 100, it is replaced with NaN
# they year columns are the 3rd column onwards
df_agriculture_outliers_removed.iloc[:, 2:] = df_agriculture_outliers_removed.iloc[:, 2:].map(
    lambda v: v if v >= 0 and v<= 100 else None
)

# creating a column to store the mean of all values in the year columns
df_agriculture_outliers_removed['Mean'] = df_agriculture_outliers_removed.iloc[:, 2:].mean(
    axis=1
)

# creating a column to store the standard deviation of all values in the year columns
# accessing the year columns from the 2nd till 2nd last column (the last column now stores the mean)
df_agriculture_outliers_removed['Standard Deviation'] = df_agriculture_outliers_removed.iloc[:, 2:-1].std(
    axis=1
)

# this function checks if an input value is an outlier
# it takes a value, mean value, standard deviation and a number 'n' as input
# it returns true if the value lies more than 'n' standard deviations away from the mean
def checkOutlier(v, mean, std, n):
    # if the value is more than 'n' standard deviations away from the mean
    if v < mean - n * std or v > mean + n * std:
        return True

# this function replaces an outlier value
# it takes the outlier's row and column number as input
# it also takes the mean value and index of the first year column as input
# it replaces the outlier value with the previous value in the row
# if the outlier lies in the first year column, it is replaced with the mean value
def replaceOutlier(row, columnNumber, mean, first):
    # if the outlier lies in the first year column (2005)
    if columnNumber == first:
        # replacing the outlier with the mean value
        row.iloc[columnNumber] = mean
    # if the outlier is in a year after 2005
    else:
        # replacing the outlier with the value before it
        row.iloc[columnNumber] = row.iloc[columnNumber-1]

# This function takes the outlier's country, indicator and year as input
# and creates an HTML unordered list to display these values
def printOutlier(country, indicator, year):
     display(HTML(f"""
         <ul>
            <li>
                <b>Country: </b>
                {country}
```

```
            </li>
            <li>
                <b>Indicator: </b>
                {indicator}
            </li>
            <li>
                <b>Year: </b>
                {year}
            </li>
        </ul>
    """))

# using markdown to print a bold heading for the outliers
display(HTML(f"""<b>Outliers replaced:</b>"""))

# Iterating over all the rows
for i, row in df_agriculture_outliers_removed.iterrows():

    # Iterating over all the year columns (3rd to 3rd last)
    for columnNumber in range(2, len(df_agriculture_outliers_removed.columns)-2):

        # checking if the value at the current row and column is an outlier
        if checkOutlier (
            row.iloc[columnNumber], # value at the current row and column
            row['Mean'], # mean value for the current row
            row['Standard Deviation'], # standard deviation for the current row
            3 # setting the number of standard deviations between the mean value and outliers
            ) == True:

            # replacing the outlier with the previous value in the row
            # if the outlier lies in the first year column, it is replaced with the mean
            replaceOutlier(
                row, # row of the outlier
                columnNumber, # column number of the outlier
                row['Mean'], # mean value for the row
                2 # index of the first year column
            )

            # printing an HTML list to represent the outlier,
            # containing the country name, indicator name and year
            printOutlier(
                row.loc['Country Name'], # country name for the current row
                row.loc['Series Name'], # indicator name for the current row
                df_agriculture_outliers_removed.columns[columnNumber] # year (column name)
            )

# removing the standard deviation column as it is no longer needed
# after outliers have been detected and replaced
df_agriculture_outliers_removed = df_agriculture_outliers_removed.drop (
    ['Standard Deviation'], # standard deviation column
    axis=1 #indicating that a column must be removed, not a row
)

# recalculating the mean for each row after the outliers have been replaced
df_agriculture_mean = df_agriculture_outliers_removed.copy(deep = True)
df_agriculture_mean['Mean'] = df_agriculture_mean.iloc[:, 2:-1].mean(axis=1)

# creating a new dataframe so that previous pipline steps are preserved
df_agriculture_years_removed = df_agriculture_mean.copy(deep = True)

# After storing the mean value of all years,
# removing the year columns as they are unneccesary for the purpose of this analysis
df_agriculture_years_removed = df_agriculture_years_removed.drop (
    df_agriculture_outliers_removed.columns[2:-1], # year columns (3rd till 2nd last)
    axis=1 # column axis
)
#displaying the first few rows of the dataframe to view the result
df_agriculture_years_removed.head()
```

Outliers replaced:

- **Country:** Armenia
- **Indicator:** Employment in agriculture (% of total employment) (modeled ILO estimate)
- **Year:** 2019 [YR2019]

- **Country:** Azerbaijan
- **Indicator:** Agriculture, forestry, and fishing, value added (% of GDP)
- **Year:** 2005 [YR2005]

- **Country:** Barbados
- **Indicator:** Agriculture, forestry, and fishing, value added (% of GDP)
- **Year:** 2020 [YR2020]

- **Country:** Brunei Darussalam
- **Indicator:** Employment in agriculture (% of total employment) (modeled ILO estimate)
- **Year:** 2019 [YR2019]

- **Country:** Caribbean small states
- **Indicator:** Agriculture, forestry, and fishing, value added (% of GDP)
- **Year:** 2005 [YR2005]

- **Country:** Georgia
- **Indicator:** Agriculture, forestry, and fishing, value added (% of GDP)
- **Year:** 2005 [YR2005]

- **Country:** Latvia
- **Indicator:** Agriculture, forestry, and fishing, value added (% of GDP)
- **Year:** 2022 [YR2022]

- **Country:** Puerto Rico
- **Indicator:** Employment in agriculture (% of total employment) (modeled ILO estimate)
- **Year:** 2005 [YR2005]

- **Country:** Rwanda
- **Indicator:** Agriculture, forestry, and fishing, value added (% of GDP)
- **Year:** 2005 [YR2005]

- **Country:** Spain
- **Indicator:** Employment in agriculture (% of total employment) (modeled ILO estimate)
- **Year:** 2005 [YR2005]

Out[4]:

	Country Name	Series Name	Mean
0	Afghanistan	Employment in agriculture (% of total employment) (modeled ILO estimate)	50.538306
1	Afghanistan	Agriculture, forestry, and fishing, value added (% of GDP)	27.144481
2	Africa Eastern and Southern	Employment in agriculture (% of total employment) (modeled ILO estimate)	59.294122
3	Africa Eastern and Southern	Agriculture, forestry, and fishing, value added (% of GDP)	12.071447
4	Africa Western and Central	Employment in agriculture (% of total employment) (modeled ILO estimate)	48.516329

Structuring Agricultural Data For Purpose

Seperating employment(%) in agriculture and GDP(%) from agriculture, forestry and fishing

The agriculture dataset contains data about two indicators:

- **Employment in agriculture (% of total employment)**
- **Agriculture, forestry, and fishing, value added (% of GDP)**

The data on each indicator will be seperated into seperate dataframes

In [5]:
```
# this function takes a dataframe and indicator as input
# it returns a seperated dataframe for the input indicator
def separateIndicator(df_combined, indicator):
    # creating a seperate dataframe for the indicator
    df = df_combined[df_combined['Series Name'] == indicator]
    # removing the indicator name column as the new dataframe contains only the one indicator
    df = df.drop('Series Name', axis=1)
    # setting the index to the country name, as this can uniquely identify each row
    df = df.set_index('Country Name')
    return df

# This function takes two seperate input dataframes for the indicators
# agricultural employment(%) and agricultural GDP(%) respectively.
# It uses HTML to display side by side the first few rows of each dataframe,
# along with their respective indicator names and number of rows.
def displaySeperateAgricultureTables(employment, GDP):
    display(HTML(f"""
    <table style="width: 100%;">
        <thead>
            <tr>
                <th style="border: 2px solid; background-color: lightblue">
                    Employment in agriculture (% of total employment)
                </th>
                <th style="border: 2px solid; background-color: lightblue">
                    Agriculture, forestry, and fishing, value added (% of GDP)
                </th>
            </tr>
        </thead>
        <tbody>
            <tr>
                <td style="border: 2px solid;">
                    {employment.head().to_html()}
                    <b>Rows: {len(employment)}</b>
                </td>
                <td style="border: 2px solid;">
                    {GDP.head().to_html()}
                    <b>Rows: {len(GDP)}</b>
                </td>
            </tr>
        </tbody>
    </table>
    """))

# creating a seperate dataframe for the indicator agriculture employment percentage
df_agriculture_employment = separateIndicator (
    df_agriculture_years_removed, # combined dataframe
    'Employment in agriculture (% of total employment) (modeled ILO estimate)' # indicator name
)

# creating a seperate dataframe for the indicator agriculture, forestry, and fishing GDP percentage
df_agriculture_GDP = separateIndicator (
    df_agriculture_years_removed, # combined dataframe
    'Agriculture, forestry, and fishing, value added (% of GDP)' # indicator name
)

# Using HTML to display the first few rows of the seperate dataframes,
# and print their respective indicator titles and number of rows
displaySeperateAgricultureTables (
    df_agriculture_employment, # agricultural employment indicator dataframe
    df_agriculture_GDP # agricultural GDP indicator dataframe
)
```

Employment in agriculture (% of total employment)	
	Mean
Country Name	
Afghanistan	50.538306
Africa Eastern and Southern	59.294122
Africa Western and Central	48.516329
Albania	41.517132
Algeria	11.857264

Rows: 230

Agriculture, forestry, and fishing, value added (% of GDP)	
	Mean
Country Name	
Afghanistan	27.144481
Africa Eastern and Southern	12.071447
Africa Western and Central	21.287569
Albania	18.230432
Algeria	9.321431

Rows: 230

Filtering Countries with a High Percentage of Employment in agriculture

This analysis explores the effect of GDP(%) of agriculture, forestry and fishing on various education completion disparities in countries with a high percentage of employment in agriculture. These countries are filtered out through the following steps:

- The dataframe for employment in agriculture is filtered to contain only those countries where the mean employment is greater than 35%
- These countries are stored in a list, which will be used when handling education data later

In [6]:
```
# creating a new dataframe to store the agricultural employment(%) data for only those countries
# that have a mean agricultural employment greater than 35%
df_agriculture_employment_high = df_agriculture_employment[df_agriculture_employment['Mean'] > 35]

# storing country names in a list
countries = df_agriculture_employment_high.index.tolist()

# using HTML to display the number of countries with agricultural employment greater than 35%,
# as well as print the list of countries and the filtered dataframe
display(HTML(f"""
    <b>{len(df_agriculture_employment_high)} countries with agricultural employment greater than 35%</b><br>

    <!-- printing the list of country names -->
    {countries}

    <!-- printing the filtered table with agricultural employment greater than 35% -->
    {df_agriculture_employment_high.to_html()}
"""))
```

78 countries with agricultural employment greater than 35%

['Afghanistan', 'Africa Eastern and Southern', 'Africa Western and Central', 'Albania', 'Angola', 'Armenia', 'Azerbaijan', 'Bangladesh', 'Benin', 'Bhutan', 'Burkina Faso', 'Burundi', 'Cambodia', 'Cameroon', 'Central African Republic', 'Chad', 'Comoros', 'Congo, Dem. Rep.', "Cote d'Ivoire", 'Early-demographic dividend', 'Equatorial Guinea', 'Eritrea', 'Ethiopia', 'Fragile and conflict affected situations', 'Gambia, The', 'Georgia', 'Ghana', 'Guinea', 'Guinea-Bissau', 'Haiti', 'Heavily indebted poor countries (HIPC)', 'IDA & IBRD total', 'IDA blend', 'IDA only', 'IDA total', 'India', 'Indonesia', 'Kenya', 'Lao PDR', 'Least developed countries: UN classification', 'Liberia', 'Low & middle income', 'Low income', 'Lower middle income', 'Madagascar', 'Malawi', 'Mali', 'Mauritania', 'Moldova', 'Morocco', 'Mozambique', 'Myanmar', 'Nepal', 'Niger', 'Nigeria', 'Pacific island small states', 'Pakistan', 'Pre-demographic dividend', 'Rwanda', 'Sierra Leone', 'Solomon Islands', 'South Asia', 'South Asia (IDA & IBRD)', 'South Sudan', 'Sub-Saharan Africa', 'Sub-Saharan Africa (excluding high income)', 'Sub-Saharan Africa (IDA & IBRD countries)', 'Sudan', 'Tajikistan', 'Tanzania', 'Thailand', 'Timor-Leste', 'Togo', 'Uganda', 'Vanuatu', 'Viet Nam', 'Zambia', 'Zimbabwe']

	Mean
Country Name	
Afghanistan	50.538306
Africa Eastern and Southern	59.294122
Africa Western and Central	48.516329
Albania	41.517132
Angola	51.336414
Armenia	52.354946
Azerbaijan	37.343866
Bangladesh	43.643498
Benin	38.528912
Bhutan	56.326378
Burkina Faso	77.509423
Burundi	87.249384
Cambodia	47.891763
Cameroon	50.342578
Central African Republic	72.959467
Chad	72.854986
Comoros	41.79604
Congo, Dem. Rep.	62.45302
Cote d'Ivoire	46.326733
Early-demographic dividend	38.781405
Equatorial Guinea	55.335441
Eritrea	66.225319
Ethiopia	70.535303
Fragile and conflict affected situations	49.224966
Gambia, The	52.721113
Georgia	46.151424
Ghana	43.781898
Guinea	64.495841
Guinea-Bissau	55.297846

Country Name	Mean
Haiti	48.090645
Heavily indebted poor countries (HIPC)	61.438541
IDA & IBRD total	35.825008
IDA blend	40.290395
IDA only	56.452597
IDA total	51.318556
India	46.886999
Indonesia	35.128573
Kenya	37.460842
Lao PDR	68.793453
Least developed countries: UN classification	58.980515
Liberia	44.352338
Low & middle income	37.194886
Low income	62.866514
Lower middle income	44.172342
Madagascar	73.883349
Malawi	66.144746
Mali	66.846905
Mauritania	35.861376
Moldova	55.283505
Morocco	37.304126
Mozambique	74.147248
Myanmar	51.830338
Nepal	65.843239
Niger	74.542356
Nigeria	40.156553
Pacific island small states	36.842067
Pakistan	41.263953
Pre-demographic dividend	54.289375
Rwanda	68.100441
Sierra Leone	56.575354
Solomon Islands	41.673955
South Asia	45.970535
South Asia (IDA & IBRD)	45.970535
South Sudan	61.309366
Sub-Saharan Africa	55.151003
Sub-Saharan Africa (excluding high income)	55.151003

Country Name	Mean
Sub-Saharan Africa (IDA & IBRD countries)	55.151003
Sudan	43.766122
Tajikistan	49.53305
Tanzania	68.835927
Thailand	36.556783
Timor-Leste	51.972068
Togo	38.984228
Uganda	70.062737
Vanuatu	56.382709
Viet Nam	43.51171
Zambia	63.292729
Zimbabwe	65.595862

Processing Education Dataset

Loading Education Dataset

In [7]:

```
# creating a dataframe to store the education data from the csv file
df_education = pandas.read_csv("archive/education_data.csv")

# displaying the dataframe ot assess its structure
df_education
```

Out[7]:

	Country Name	Country Code	Series	Series Code	2005 [YR2005]	2006 [YR2006]	2007 [YR2007]	2008 [YR2008]	2009 [YR2009]	2 [YR2(
0	Afghanistan	AFG	Completion rate, lower secondary education, rural, fourth quintile, female (%)	UIS.CR.2.RUR.Q4.F	..	..	..	..	..	
1	Afghanistan	AFG	Completion rate, lower secondary education, rural, fourth quintile, male (%)	UIS.CR.2.RUR.Q4.M	..	..	..	..	..	
2	Afghanistan	AFG	Completion rate, lower secondary education, rural, middle quintile, female (%)	UIS.CR.2.RUR.Q3.F	..	..	..	..	..	
3	Afghanistan	AFG	Completion rate, lower secondary education, rural, middle quintile, male (%)	UIS.CR.2.RUR.Q3.M	..	..	..	..	..	
4	Afghanistan	AFG	Completion rate, lower secondary education, rural, poorest quintile, female (%)	UIS.CR.2.RUR.Q1.F	..	..	..	..	..	
...	...	...	...	...	...	...	...	...	...	
16320	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
16321	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
16322	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
16323	Data from database: Education Statistics - All Indicators	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
16324	Last Updated: 06/25/2024	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	

16325 rows × 23 columns

Cleaning Education Dataset

The following observations can be made from the dataframe displayed above

Structure

<table>
<tr><th>Rows</th><th>Columns</th></tr>
<tr><td>For each country, there are are some rows for education data based on variations in a set of categories,. The set of categories are specified as comma seperated strings in the 'Series' column. These series categories were selected and downloaded from the dataset, according to the focus of this analysis. The set of categories contain:

1. Education metric
A. Completion rate
2. Education level
A. Primary
B. Lower secondary
C. Upper secondary
3. Region
A. Urban
B. Rural
4. Financial status (quintile)
A. Poorest quintile
B. Second quintile
C. Middle quintile
D. Fourth quintile
E. Richest quintile
5. Gender
A. Male
B. Female</td>
<td>The following columns can be observed in the dataframe displayed above

1. Country Name
2. Country Code
3. Series
The comma seperated strings containing the set of five categories:
metric,
education level,
region,
financial quintile
gender
4. Series Code
5. Years from 2005 till 2023</td></tr>
</table>

Initial cleaning steps

After viewing the dataframe, I outlined the following steps to clean and validate the data

1. The year columns represent numerical values. However, some values are not numeric. For example, the string '..' is used to represent empty values. Therefore, all values under the year columns must be coerced to numerical values to ensure consistency
2. There are unnecessary rows towards the end of the dataframe, which contain metadata or NaN values. These rows must be deleted
3. If an entire row contains only NaN values in the year columns, it must be deleted as it is unusable
4. The dataframe contains all countries. For this exploriation, only the countries with a high percentage of agricultural employment will be analyzed. Therefore, the rows must be filtered to contain only those countries (stored previoused in the list 'countries')
5. Series codes and country codes are unnecessary to this analysis. Therefore, these columns must be deleted for simplicity
6. Each string in the series column represents a combination of five comma seperated categories. Therefore, all the values in the series column must have exactly 4 commas
7. If a row has the same country name and series name (combination of categories), that row is a duplicate and should be removed
8. For each country, there must be 60 rows. Each row represents a different combination of the following categories:
    - 1 metric
    - 3 education levels
    - 2 regions
    - 5 financial quintiles
    - 2 genders

For this analysis, all categories are used in correspondance. For a certain country, if there is no data for a certain combination of categories, the data for the other categories is also unusable. Hence, countries which do not have 60 rows in the dataframe should be removed.

In [8]:
```python
# creating a new dataframe to store previous pipeline steps
df_education_numeric = df_education.copy(deep = True)

# converting all values under the year columns to numeric
# the year columns are accessed from the 5th column onwards
df_education_numeric.iloc[:, 4:] = df_education_numeric.iloc[:, 4:].apply(
    pandas.to_numeric,
    errors='coerce'
)

# removing all rows that have only NaN values in all the year columns
# this removes the unwanted rows at the end of the dataframe
# if a country has no data for one of the indicators, that row is also removed
df_education_empty_rows_removed = df_education_numeric.dropna (
    subset = df_education_numeric.columns[4:], # year columns (fifth column onwards)
    how = 'all' # removing the row if all the values are NaN
)

# filtering the education dataframe so it only contains those countries
# that have agricultural employment greater than 35% (stored in the list countries)
df_education_agricultural_countries = df_education_empty_rows_removed[
df_education_empty_rows_removed['Country Name'].isin(countries)]

#removing the unnecessary series code and country code columns
df_education_codes_removed = df_education_agricultural_countries.drop (
    ['Series Code', 'Country Code'], # series code and country code columns
    axis=1, # column axis
)

# The strings in the series column store 5 pieces of information, seperated by commas:
# filtering the dataframe to ensure that the string in the series column has 4 commas
df_education_filtered_categories = df_education_codes_removed[
df_education_codes_removed['Series'].str.count(',') == 4]

# removing any duplicate rows (with the same country name and series name)
df_education_duplicates_removed = df_education_filtered_categories.drop_duplicates (
    subset=['Country Name', 'Series'] # rows with the same country name and series name
)

# filtering the dataframe so that there are 60 rows (one for each combination of categories)
# representing each country
df_education_all_categories = df_education_duplicates_removed.groupby('Country Name').filter(
    lambda c: len(c) == 60
)

#after dropping rows, resetting the index values so they appear from 0 in sequence
df_education_all_categories.reset_index(drop=True, inplace=True)

df_education_all_categories
```

Out[8]:

	Country Name	Series	2005 [YR2005]	2006 [YR2006]	2007 [YR2007]	2008 [YR2008]	2009 [YR2009]	2010 [YR2010]	2011 [YR2011]	2012 [YR2012]	...
0	Afghanistan	Completion rate, lower secondary education, rural, fourth quintile, female (%)	NaN	NaN	NaN	NaN	NaN	NaN	12.88914	NaN	...
1	Afghanistan	Completion rate, lower secondary education, rural, fourth quintile, male (%)	NaN	NaN	NaN	NaN	NaN	NaN	42.80005	NaN	...
2	Afghanistan	Completion rate, lower secondary education, rural, middle quintile, female (%)	NaN	NaN	NaN	NaN	NaN	NaN	3.51711	NaN	...
3	Afghanistan	Completion rate, lower secondary education, rural, middle quintile, male (%)	NaN	NaN	NaN	NaN	NaN	NaN	31.15588	NaN	...
4	Afghanistan	Completion rate, lower secondary education, rural, poorest quintile, female (%)	NaN	NaN	NaN	NaN	NaN	NaN	0.03164	NaN	...
...	...	...	...	...	...	...	...	...	...	...	...
3175	Zimbabwe	Completion rate, upper secondary education, urban, poorest quintile, male (%)	0.0	NaN	NaN	NaN	NaN	0.0	NaN	NaN	...
3176	Zimbabwe	Completion rate, upper secondary education, urban, richest quintile, female (%)	10.02479	NaN	NaN	NaN	NaN	27.67635	NaN	NaN	...
3177	Zimbabwe	Completion rate, upper secondary education,	18.36383	NaN	NaN	NaN	NaN	39.84425	NaN	NaN	...

	Country Name	Series	2005 [YR2005]	2006 [YR2006]	2007 [YR2007]	2008 [YR2008]	2009 [YR2009]	2010 [YR2010]	2011 [YR2011]	2012 [YR2012]	...
		urban, richest quintile, male (%)									
3178	Zimbabwe	Completion rate, upper secondary education, urban, second quintile, female (%)	0.0	NaN	NaN	NaN	NaN	0.0	NaN	NaN	...
3179	Zimbabwe	Completion rate, upper secondary education, urban, second quintile, male (%)	0.0	NaN	NaN	NaN	NaN	0.0	NaN	NaN	...

3180 rows × 21 columns

Replacing Outliers and Storing Mean Values

The following steps are taken to replace outliers and calculate the mean values

- As the values are percentages, they must lie between 0 and 100. If a value under the year columns does not lie within this range, it ise replaced with NaN
- A new column is created to store the mean of all the year values of each row
- The year columns are removed after calculating the mean, as they are no longer required

Note: in this dataframe, outliers are not calculated using standard deviations (like how they were calculated for the agriculture dataframe). After viewing the structure of the dataframe, it can be observed that most rows contain only 1 or 2 values in the year columns. Most of the year columns are empty. Therefore, calculating outliers using standard deviations is unfeasable.

In [9]:

```
# creating a new dataframe to store education data with outliers replaced,
# so that the previous steps in the pipeline are preserved
df_education_outliers_removed = df_education_all_categories.copy(deep = True)

# if a value under the year columns is lesser than 0 or greater than 100, it is replaced with NaN
# the year columns are the 3rd column onwards
df_education_outliers_removed.iloc[:, 2:] = df_education_outliers_removed.iloc[:, 2:].map(
    lambda v: v if v >= 0 and v<= 100 else None
)

# creating a new dataframe to store education data with the mean,
# so that previous pipeline steps are preserved
df_education_mean = df_education_outliers_removed.copy(deep = True)

# creating a column to store the mean of all values in the year columns
df_education_mean['Mean'] = df_education_mean.iloc[:, 2:].mean(axis=1)

# After storing the mean of all years, removing the year columns as they are not required
# for this analysis
df_education_years_removed = df_education_mean.drop(
    df_education_mean.columns[2:-1], # year columns (third to second last)
    axis=1 # column axis
)
```

```python
#after dropping rows, resetting the index values so they appear from 0 in sequence
df_education_years_removed.reset_index(drop=True, inplace=True)

df_education_years_removed
```

Out[9]:

	Country Name	Series	Mean
0	Afghanistan	Completion rate, lower secondary education, rural, fourth quintile, female (%)	19.63095
1	Afghanistan	Completion rate, lower secondary education, rural, fourth quintile, male (%)	47.237345
2	Afghanistan	Completion rate, lower secondary education, rural, middle quintile, female (%)	8.32381
3	Afghanistan	Completion rate, lower secondary education, rural, middle quintile, male (%)	37.0882
4	Afghanistan	Completion rate, lower secondary education, rural, poorest quintile, female (%)	6.45724
...	...	...	...
3175	Zimbabwe	Completion rate, upper secondary education, urban, poorest quintile, male (%)	0.0
3176	Zimbabwe	Completion rate, upper secondary education, urban, richest quintile, female (%)	24.746802
3177	Zimbabwe	Completion rate, upper secondary education, urban, richest quintile, male (%)	34.931065
3178	Zimbabwe	Completion rate, upper secondary education, urban, second quintile, female (%)	0.0
3179	Zimbabwe	Completion rate, upper secondary education, urban, second quintile, male (%)	0.0

3180 rows × 3 columns

Structuring Education Data for Purpose

Categories

The strings in the 'Series' column contain a set of comma seperated categories. These series categories were selected and downloaded from the dataset, according to the focus of this analysis. The set of categories contain:

1. **Education metric**
   A. Completion rate
2. **Education level**
   A. Primary
   B. Lower secondary
   C. Upper secondary
3. **Region**
   A. Urban
   B. Rural
4. **Financial status (quintile)**
   A. Poorest quintile
   B. Second quintile
   C. Middle quintile
   D. Fourth quintile
   E. Richest quintile
5. **Gender**
   A. Male
   B. Female

Seperating categories

These categories will be seperated so that they can be used in the analysis. The following steps are taken to store the categories:

- The strings in the 'Series' column should be split into substrings by a comma delimiter

- These substrings should be stored in appropriate columns representing the corresponding category
- The last substring representing the gender should be split further by a bracket '(' to contain only the portion before the bracket. This is because the substring after the last comma contains the substring '(%)' at the end
- To clean and validate the data further, the columns must be filtered to contain only the specified categories which will be used in this analysis
- After validating the data, the following unnecessary columns should be deleted:
  - **Series:** After splitting the series column into categories, it is no longer required
  - **Metric:** For the purpose of this analysis, only the education completion metric is used. After validating that all rows contain data about this metric, the column is not required anymore

In [10]:
```
# creating a new dataframe to store columns for the seperate categories,
# so that the previous pipeline steps are preserved
df_education_categories = df_education_years_removed.copy(deep = True)

# storing the first part of the series strings in a column 'Metric'
df_education_categories['Metric'] = df_education_categories['Series'].str.split(',').str[0]

# storing the second part of the series strings in a column 'Level'
df_education_categories['Level'] = df_education_categories['Series'].str.split(',').str[1]

# storing the third part of the series strings in a column 'Region'
df_education_categories['Region'] = df_education_categories['Series'].str.split(',').str[2]

# storing the fourth part of the series strings in a column 'Financial Quintile'
df_education_categories['Financial Quintile'] = df_education_categories['Series'].str.split(',').str[3]

# storing the last part of the series strings in a column 'Gender'
df_education_categories['Gender'] = df_education_categories['Series'].str.split(',').str[4]

# reducing the gender string to the part before the first bracket (before '(%)')
df_education_categories['Gender'] = df_education_categories['Gender'].str.split('(').str[0]

# filtering the dataframe to ensure it includes only the specified categories
df_education_categories = df_education_categories[df_education_categories['Metric'].isin(
    ['Completion rate']
)]
df_education_categories = df_education_categories[df_education_categories['Level'].isin(
    [' primary education',
     ' lower secondary education',
     ' upper secondary education']
)]
df_education_categories = df_education_categories[df_education_categories['Region'].isin(
    [' urban',
     ' rural']
)]
df_education_categories = df_education_categories[df_education_categories['Financial Quintile'].isin(
    [' richest quintile',
     ' middle quintile',
     ' second quintile',
     ' fourth quintile',
     ' poorest quintile']
)]
df_education_categories = df_education_categories[df_education_categories['Gender'].isin(
    [' female ',
     ' male ']
)]

# for the purpose of this analysis, only one education metric is used: completion rate
# after validating that the metric for all rows is the completion rate, the metric column can be deleted
df_education_categories = df_education_categories.drop(
    ['Metric', 'Series'], # metric and series columns
    axis = 1 # column axis
)

df_education_categories
```

Out[10]:

	Country Name	Mean	Level	Region	Financial Quintile	Gender
0	Afghanistan	19.63095	lower secondary education	rural	fourth quintile	female
1	Afghanistan	47.237345	lower secondary education	rural	fourth quintile	male
2	Afghanistan	8.32381	lower secondary education	rural	middle quintile	female
3	Afghanistan	37.0882	lower secondary education	rural	middle quintile	male
4	Afghanistan	6.45724	lower secondary education	rural	poorest quintile	female
...	...	...	...	...	...	...
3175	Zimbabwe	0.0	upper secondary education	urban	poorest quintile	male
3176	Zimbabwe	24.746802	upper secondary education	urban	richest quintile	female
3177	Zimbabwe	34.931065	upper secondary education	urban	richest quintile	male
3178	Zimbabwe	0.0	upper secondary education	urban	second quintile	female
3179	Zimbabwe	0.0	upper secondary education	urban	second quintile	male

3180 rows × 6 columns

Seperating Education Levels

The education dataframe contains data for 3 education levels.

1. **Primary Education**
2. **Lower Secondary Education**
3. **Upper Secondary Education**

To seperate the education levels, the following steps are taken

- A dictionary is created to store a filtered dataframe for each education level
- The 'Level' column is deleted from the filtered level dataframes

In [11]:

```
# creating an empty dictionary to store education levels
df_education_levels = {}

# storing filtered dataframes for each level in the dictionary
df_education_levels['Primary'] = df_education_categories[
df_education_categories['Level'] == ' primary education'
]
df_education_levels['Lower Secondary'] = df_education_categories[
df_education_categories['Level'] == ' lower secondary education'
]
df_education_levels['Upper Secondary'] = df_education_categories[
df_education_categories['Level'] == ' upper secondary education'
]

# iterating over all the dictionary items (seperate education level dataframes)
for level, df_education_level in df_education_levels.items():
    # deleting the level column as it is no longer required
    df_education_levels[level] = df_education_level.drop(
        'Level', # level column
        axis = 1, # column axis
    )

    # reordering the columns so that the mean appears as the last column
    df_education_levels[level] = df_education_levels[level][['Country Name',
                                                               'Region',
                                                               'Financial Quintile',
                                                               'Gender',
                                                               'Mean']]

    # displaying the level and its corresponding dataframe
```

```
    display(HTML(f""" <b>{level} Education</b>"""))
    display(df_education_levels[level])
```

Primary Education

	Country Name	Region	Financial Quintile	Gender	Mean
20	Afghanistan	rural	fourth quintile	female	34.90063
21	Afghanistan	rural	fourth quintile	male	70.46655
22	Afghanistan	rural	middle quintile	female	21.055855
23	Afghanistan	rural	middle quintile	male	53.91436
24	Afghanistan	rural	poorest quintile	female	19.39543
...	...	...	...	...	...
3155	Zimbabwe	urban	poorest quintile	male	0.0
3156	Zimbabwe	urban	richest quintile	female	97.70333
3157	Zimbabwe	urban	richest quintile	male	98.103547
3158	Zimbabwe	urban	second quintile	female	0.0
3159	Zimbabwe	urban	second quintile	male	0.0

1060 rows × 5 columns

Lower Secondary Education

	Country Name	Region	Financial Quintile	Gender	Mean
0	Afghanistan	rural	fourth quintile	female	19.63095
1	Afghanistan	rural	fourth quintile	male	47.237345
2	Afghanistan	rural	middle quintile	female	8.32381
3	Afghanistan	rural	middle quintile	male	37.0882
4	Afghanistan	rural	poorest quintile	female	6.45724
...	...	...	...	...	...
3135	Zimbabwe	urban	poorest quintile	male	0.0
3136	Zimbabwe	urban	richest quintile	female	87.769543
3137	Zimbabwe	urban	richest quintile	male	92.571247
3138	Zimbabwe	urban	second quintile	female	0.0
3139	Zimbabwe	urban	second quintile	male	0.0

1060 rows × 5 columns

Upper Secondary Education

	Country Name	Region	Financial Quintile	Gender	Mean
40	Afghanistan	rural	fourth quintile	female	7.51374
41	Afghanistan	rural	fourth quintile	male	32.64284
42	Afghanistan	rural	middle quintile	female	3.828965
43	Afghanistan	rural	middle quintile	male	19.32065
44	Afghanistan	rural	poorest quintile	female	2.67479
...	...	...	...	...	...
3175	Zimbabwe	urban	poorest quintile	male	0.0
3176	Zimbabwe	urban	richest quintile	female	24.746802
3177	Zimbabwe	urban	richest quintile	male	34.931065
3178	Zimbabwe	urban	second quintile	female	0.0
3179	Zimbabwe	urban	second quintile	male	0.0

1060 rows × 5 columns

Seperating Disparities

The education dataframe contains data for 3 disparities.

1. **Financial Disparity**
2. **Gender Disparity**
3. **Region (Urban/Rural) Disparity**

In the previous step, dataframes for each educational level (primary, lower secondary, upper secondary) were stored in a dictionary. Now, that dictionary will be copied, and a nested dictionary will be created within each level. The nested dictionary will contain filtered dataframes for each disparity.

In [12]:

```
# creating a copy of the dictionary that stores dataframes for each educational level
# the new copy will contain a nested dictionary within each level
# the nested dictionary will store seperate dataframes for each disparity
df_education_levels_disparities = df_education_levels.copy()

# this function takes an education level dataframe and the column name of a disparity as input
# it returns a filtered dataframe containing data for the input disparity
def seperateDisparity(df, disparity):
    #creating a copy of the input dataframe
    df_disparity = df.copy(deep = True)

    # filtering the dataframe to contain only 3 columns,
    # for the country name, the disparity and the mean value
    df_disparity = df_disparity.loc[:, ['Country Name', 'Mean', disparity]]

    # storing a single mean value for rows that have the same disparity category and country name,
    # (the mean of all the values in the 'Mean' column)
    df_disparity = df_disparity.groupby(['Country Name', disparity], as_index = False)['Mean'].mean()

    # reshaping the dataframe so it contains seperate columns for each disparity category
    df_disparity = df_disparity.pivot (
        index='Country Name', # setting the index to the country name
        columns = disparity, # creating seperate columns for the disparity categories
        values='Mean' # storing the values from the mean column under the disparity categories columns
    )

    # accessing the agriculture GDP (%) from the dataframe df_agriculture_GDP created previously
    # storing the GDP in a seperate column
    df_disparity['Agriculture, Forestry, and Fishing GDP Mean (%)'] = df_agriculture_GDP['Mean']

    # ensuring that the GDP contains numerical values
```

```
    df_disparity['Agriculture, Forestry, and Fishing GDP Mean (%)'] = pandas.to_numeric(
        df_disparity['Agriculture, Forestry, and Fishing GDP Mean (%)'],
        errors='coerce'
    )

    # creating a seperate column to represent the education disparity
    # the standard deviation of all the values in the disparity categories
    # is used to represent the disparity
    df_disparity['Disparity'] = df_disparity.iloc[:, :-1].std(axis=1)

    # ensuring that the disparity contains numerical values
    df_disparity['Disparity'] = pandas.to_numeric(
        df_disparity['Disparity'],
        errors='coerce'
    )

    # returning the seperated dataframe for the disparity
    return df_disparity

# iterating over all the education level dataframes in the dictionary
for level, df_education_level in df_education_levels.items():
    # creating an empty nested dictionary within a copy of the level
    # this dictionary will store dataframes for each disparity
    df_education_levels_disparities[level] = {}

    # storing seperated dataframes for financial disparities at each level
    df_education_levels_disparities[level]['Financial'] = seperateDisparity (
        df_education_level, # the dataframe which will be filtered to get seperate disparity dataframe
        "Financial Quintile" # the disparity
    )

    # storing seperated dataframes for gender disparities at each level
    df_education_levels_disparities[level]['Gender'] = seperateDisparity (
        df_education_level, # the dataframe which will be filtered to get seperate disparity dataframe
        "Gender" # the disparity
    )

    # storing seperated dataframes for regional disparities at each level
    df_education_levels_disparities[level]['Region'] = seperateDisparity (
        df_education_level, # the dataframe which will be filtered to get seperate disparity dataframe
        "Region" # the disparity
    )

    # reordering the columns of the financial disparity dataframes
    # so that the financial quintiles appear in order from poorest to richest
    df_education_levels_disparities[level]['Financial']=df_education_levels_disparities[level]['Financial'][[
        ' poorest quintile',
        ' second quintile',
          ' middle quintile',
          ' fourth quintile',
          ' richest quintile',
          'Agriculture, Forestry, and Fishing GDP Mean (%)',
          'Disparity'
    ]]

# iterating over all the education levels in the dictionary
for level, df_education_level in df_education_levels_disparities.items():
    # printing the level
    display(HTML(f"""<b><h3>{level} Education: </h3>"""))
    # iterating over all the disparity dataframes within the education level dictionary
    for disparity, df_disparity in df_education_level.items():
        # displaying the name of the disparity
        display(HTML(f"""<b><h4>{disparity} disparities </h4><br> countries: </b>{len(df_disparity)}"""))
        #displaying the first few rows of the dataframe for the education level and disparity
        display(df_disparity.head())
```

Primary Education:

Financial disparities

countries: 53

Financial Quintile	poorest quintile	second quintile	middle quintile	fourth quintile	richest quintile	Agriculture, Forestry, and Fishing GDP Mean (%)	Disparity
Country Name							
Afghanistan	28.636681	39.153813	40.732916	54.633809	69.171392	27.144481	15.704051
Albania	71.802687	71.568993	97.246879	82.034327	49.385435	18.230432	17.469628
Angola	9.528575	31.052225	59.270633	60.9399	44.398085	8.137081	21.403062
Armenia	57.864391	91.597302	82.794827	50.0	49.953589	14.151406	19.467836
Azerbaijan	65.759735	98.935525	97.701497	97.803295	49.776955	5.891423	22.831797

Gender disparities

countries: 53

Gender	female	male	Agriculture, Forestry, and Fishing GDP Mean (%)	Disparity
Country Name				
Afghanistan	34.024201	58.907244	27.144481	17.594968
Albania	77.222872	71.592457	18.230432	3.981304
Angola	33.293653	48.782114	8.137081	10.951996
Armenia	66.478669	66.405375	14.151406	0.051827
Azerbaijan	84.924783	79.06602	5.891423	4.142771

Region disparities

countries: 53

Region	rural	urban	Agriculture, Forestry, and Fishing GDP Mean (%)	Disparity
Country Name				
Afghanistan	44.300311	48.631134	27.144481	3.062355
Albania	71.298158	77.517171	18.230432	4.397507
Angola	29.474439	52.601328	8.137081	16.353180
Armenia	53.11801	79.766033	14.151406	18.842998
Azerbaijan	77.334416	86.656387	5.891423	6.591629

Lower Secondary Education:

Financial disparities

countries: 53

Financial Quintile	poorest quintile	second quintile	middle quintile	fourth quintile	richest quintile	Agriculture, Forestry, and Fishing GDP Mean (%)	Disparity
Country Name							
Afghanistan	17.646422	21.792056	24.688737	32.58117	52.508015	27.144481	13.796205
Albania	50.136847	59.852915	81.165381	68.801739	48.09021	18.230432	13.708021
Angola	2.220708	8.13612	25.315323	32.431575	36.14	8.137081	14.971217
Armenia	49.413874	71.239653	67.164735	46.294173	47.206023	14.151406	11.952665
Azerbaijan	43.4645	87.154027	92.3358	70.918307	48.568242	5.891423	22.056840

Gender disparities

countries: 53

Gender	female	male	Agriculture, Forestry, and Fishing GDP Mean (%)	Disparity
Country Name				
Afghanistan	18.596192	41.090368	27.144481	15.905785
Albania	61.261714	61.957123	18.230432	0.491728
Angola	20.083001	21.614489	8.137081	1.082926
Armenia	57.618841	54.908542	14.151406	1.916471
Azerbaijan	70.731833	66.244518	5.891423	3.173011

Region disparities

countries: 53

Region	rural	urban	Agriculture, Forestry, and Fishing GDP Mean (%)	Disparity
Country Name				
Afghanistan	28.449098	31.237463	27.144481	1.971672
Albania	53.89201	69.326827	18.230432	10.914064
Angola	11.289939	30.407551	8.137081	13.518193
Armenia	43.170738	69.356645	14.151406	18.516232
Azerbaijan	62.365366	74.610985	5.891423	8.658960

Upper Secondary Education:

Financial disparities

countries: 53

Financial Quintile Country Name	poorest quintile	second quintile	middle quintile	fourth quintile	richest quintile	Agriculture, Forestry, and Fishing GDP Mean (%)	Disparity
Afghanistan	11.131644	10.352127	13.640796	17.897709	38.424679	27.144481	11.634283
Albania	30.939032	42.388292	59.78915	58.806809	46.728639	18.230432	12.037358
Angola	0.931987	1.970673	7.207863	10.093765	24.95817	8.137081	9.663983
Armenia	25.389453	55.062457	53.503774	36.502195	41.588678	14.151406	12.333209
Azerbaijan	26.454518	66.183245	75.18394	86.023492	44.680527	5.891423	24.008111

Gender disparities

countries: 53

Gender Country Name	female	male	Agriculture, Forestry, and Fishing GDP Mean (%)	Disparity
Afghanistan	10.957503	25.621279	27.144481	10.368856
Albania	46.910691	48.550078	18.230432	1.159222
Angola	7.751481	10.313502	8.137081	1.811622
Armenia	46.831866	37.986756	14.151406	6.254437
Azerbaijan	56.938896	62.471393	5.891423	3.912066

Region disparities

countries: 53

Region Country Name	rural	urban	Agriculture, Forestry, and Fishing GDP Mean (%)	Disparity
Afghanistan	17.632887	18.945895	27.144481	0.928437
Albania	37.577515	57.883254	18.230432	14.358325
Angola	2.685624	15.379359	8.137081	8.975826
Armenia	29.333597	55.485026	14.151406	18.491853
Azerbaijan	59.034663	60.375626	5.891423	0.948204

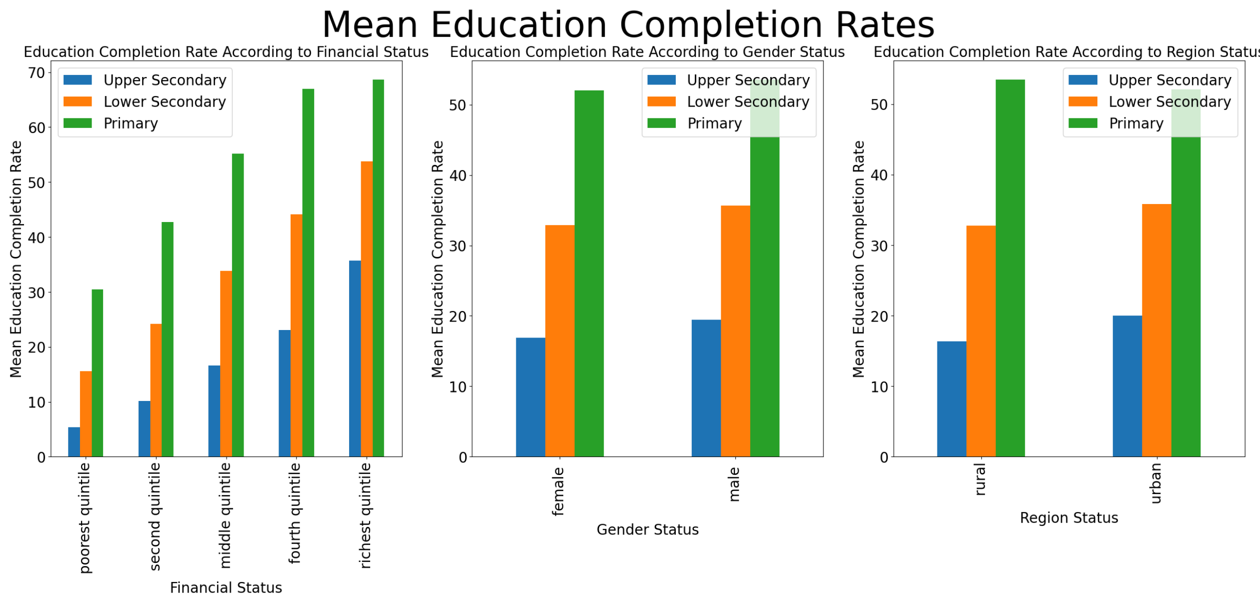
Data Visualisation and Analysis

Visualising Financial, Gender and Regional Disparities in Education Completion Across Primary, Lower Secondary and Upper Secondary Education Levels in Countries with Major Employment in Agriculture

In [13]:

```
# creating 3 subplots
# each subplot will have a bargraph for one of the disparities at each education level
f, axis = plt.subplots(1, 3, figsize=(30, 10))

# iterating over the seperated disparity dataframes in the education levels
# the purpose of specifying the primary level in this for loop is to access the
# index 'j' and the key 'disparity'
# Each education level dictionary contains the same keys for all disparities,
# therefore specifying the primary level can access those keys
```

```python
for j, (disparity, df_disparity) in enumerate(df_education_levels_disparities['Primary'].items()):

    # creating a dataframe which will store data for the disparity at each education level
    disparityGraph = pandas.DataFrame({
    'Upper Secondary': df_education_levels_disparities['Upper Secondary'][disparity].iloc[:, :-2].mean(),
    'Lower Secondary': df_education_levels_disparities['Lower Secondary'][disparity].iloc[:, :-2].mean(),
    'Primary': df_education_levels_disparities['Primary'][disparity].iloc[:, :-2].mean()
    })

    # plotting a bargraph for the disparity
    disparityGraph.plot(
        kind='bar', # bar graph
        ax = axis[j] # setting the axis of the subplot to the index
    )

    # setting the title of each subplot
    axis[j].set_title(
        'Education Completion Rate According to ' + str(disparity) + " Status",
        fontsize=20
    )

    # setting the label of the x axis of the subplot
    axis[j].set_xlabel(
        str(disparity) + " Status",
        fontsize=20
    )

    # setting the label of the y axis of the subplot
    axis[j].set_ylabel(
        'Mean Education Completion Rate',
        fontsize=20
    )

    #creating a legend to represent education levels
    axis[j].legend(fontsize = 20)

    #increasing the size of the numbers on both axis
    axis[j].tick_params(
        axis='both',
        labelsize=20
    )

#setting the title of the entire figure with all the subplots
f.suptitle (
    "Mean Education Completion Rates",
    fontsize = 50
)

plt.show()
```



Analyzing Financial, Gender and Regional Disparities in Education Completion Across Primary, Lower Secondary and Upper Secondary Education Levels in Countries with Major Employment in Agriculture

The bar graphs above visualize the education completion rates for different financial, gender and region categories across the primary, lower secondary and upper secondary education levels. The height of the bars represent the completion rates, whereas the difference between the heights of the bars represent the disparity.

Education Levels

- **Primary:** The primary education level (green) has the highest completion rates (bar heights) and the least disparity in education completion (difference between bar heights). This is probably because more children can afford to complete primary education despite their status. However, as their education progresses, they might drop out due to increased fees, or they might take up jobs etc.
- **Lower Secondary:** The lower secondary education level (orange) has the second highest completion rates. Education completion disparities are most visible between financial quintiles. From the poorest quintile (left) to the richest quintile right), the length of the bars for each education level increases. In other words, for each education level, the education completion rate changes between each quintile
- **Upper Secondary:** The upper secondary education level (blue) has the least completion rates.

This is probably because more children can afford to complete primary education despite their status. However, as their education progresses, they might drop out due to increased fees, or they might take up jobs etc.
Disparities

- **Financial:** Education completion disparities are most visible between financial quintiles. From the poorest quintile (left) to the richest quintile right), the length of the bars for each education level increases. In other words, for each education level, the education completion rate changes between each quintile
- **Gender:** For each education level, males have a higher education completion rate than females. However, this disparity is not as visible as the financial disparity
- **Region:** Urban regions have a higher education completion rate at the lower secondary and upper secondary levels. However, rural areas have a higher education completion rate at the primary level. This is probably because primary schools are of a similar standard in both urban and rural regions. However, as the education level progresses, the standard of the schools in rural areas degrades

The most education completion disparities are between financial quintiles, followed by gender, followed by region

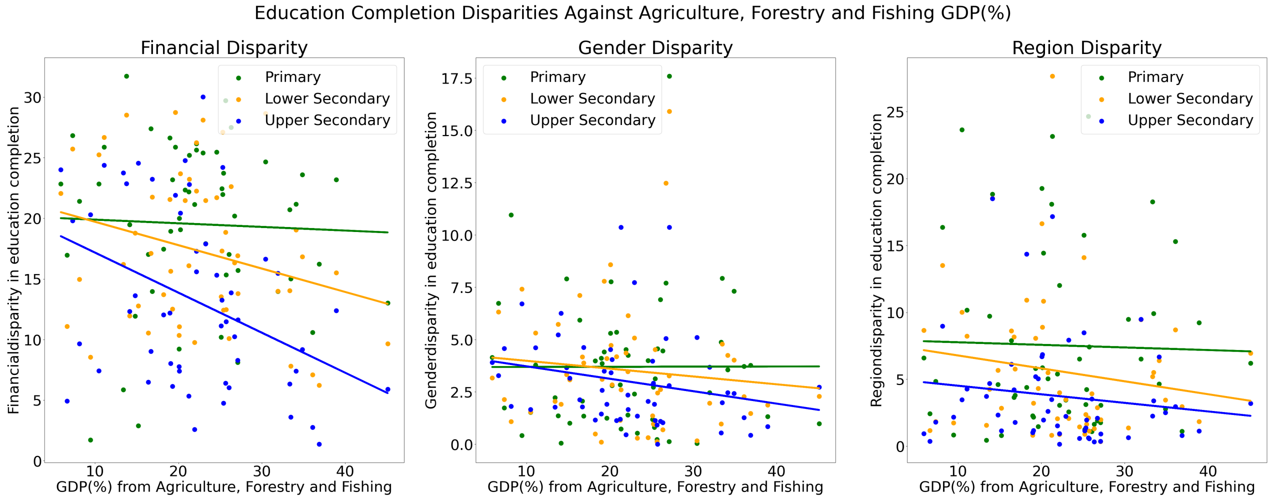
Visualising the Effect of GDP(%) from Agriculture, Forestry and Fishing on Education Completion Disparities (Financial, Gender and Region) Across

Education Levels (Primary, Lower Secondary and Upper Secondary) in Countries with Major Employment in Agriculture

In [14]:

```
f, axis = plt.subplots(1, 3, figsize=(60, 20))

# iterating over the dataframes for the three education levels (primary, lower secondary, upper secondary)
for j, (level, df_education_level) in enumerate(df_education_levels_disparities.items()):
    # iterating over the dataframes for the three disparities at each level (financial, gender, region)
    for i, (disparity, df_disparity) in enumerate(df_education_level.items()):

        # storing the x axis values from the agriculture GDP column
        GDP = df_disparity['Agriculture, Forestry, and Fishing GDP Mean (%)'].values

        # storing the y axis values from the disparity column
        Disparity = df_disparity['Disparity'].values

        # storing 3 different colours to represent 3 education levels
        levelColours = ['green', 'orange', 'blue']

        # creating a scatter plot for the given disparity
        # at the index of the disparity dataframe
        axis[i].scatter(
            GDP, # x values
            Disparity, # y values
            color=levelColours[j], # setting the colour based on the education level index
            label = level, #setting the label to the education level
            s = 150 # setting an appropriate size of scatter points
        )

        # storing the slope and intercep of the line of best fit
        slope, intercept = np.polyfit(
            GDP, # x values
            Disparity, # y values
            1
        )

        #plotting the line of best fit for each level in each subplot
        axis[i].plot (
            GDP, # x values
            GDP * slope + intercept, # line of best fit
            color = levelColours[j], # setting the colour based on the education level index
            linewidth=5 # setting an appropriate width to the line of best fit
        )

        # setting the subplot's title
        axis[i].set_title (
            str(disparity) + " Disparity",
            fontsize = 50
        )

        # setting the subplot's x-axis label
        axis[i].set_xlabel (
            'GDP(%) from Agriculture, Forestry and Fishing',
            fontsize = 40
        )

        # setting the subplot's y-axis label
        axis[i].set_ylabel (
            str(disparity) + "disparity in education completion",
            fontsize = 40
        )

        #creating a legend to represent education levels
        axis[i].legend(fontsize = 40)



        #increasing the size of the numbers on both axis
        axis[i].tick_params (
```

```
            axis='both',
            labelsize = 40
        )

#setting the title of the entire figure with all the subplots
f.suptitle(
    "Education Completion Disparities Against Agriculture, Forestry and Fishing GDP(%) ",
    fontsize = 50
)

plt.show()
```



Analyzing the Effect of GDP(%) from Agriculture, Forestry and Fishing on Education Completion Disparities (Financial, Gender and Region) Across Education Levels (Primary, Lower Secondary and Upper Secondary) in Countries with Major Employment in Agriculture

As shown in the scatter plots above, there is a clear inverse relationship between Agriculture, Forestry and Fishing GDP(%) and education completion disparities. As the Agriculture, Forestry and Fishing GDP(%), education completion disparities decrease.

Education Levels

- **Primary:** The effect of agriculture, forestry and fishing GDP(%) on education disparities at the primary level is negligable. The green lines representing the primary level do not have a visible slope. As shown in the previous bar graphs, the primary education level has the highest completion rates. Disparities appear least at this level, because more children can afford to complete primary education despite their status. As education completion rates are already high at the primary level, an increase in agricultural GDP does not affect the primary level disparities noticeably.
- **Lower Secondary:** The orange lines representing the lower secondary level have a visible downward slope. This implies that agriculture, forestry and fishing GDP(%) has an inverse relationship with education completion disparities at the lower secondary level.
- **Upper Secondary:** The blue lines representing the upper secondary level have the most downward slope on the scatter plots (except the region disparity scatter plot). As shown in the previous bar graphs, the upper secondary education level has the lowest education completion rates. Therefore, an increase in agricultural GDP affects the upper secondary level disparities significantly.

Disparities

- **Financial:** The lines on the financial disparity scatter plot have the most slope. This is because most education disparities appear between socio-economic statuses, as proven by the previous bar graph. Therefore, an increase in agricultural GDP decreases the financial disparities significantly.

- **Gender:** The lines on the financial disparity scatter plot have a downward slope. As observed in the previous bar graph, education completion disparities between genders are lesser than disparities between financial quintiles. Therefore, an increase in the agricultural GDP decreases the gender disparities, but not as much as it decreases the financial disparities.
- **Region:** The lines on the region disparity scatter plot have a downard slope. As observed in the previous bar graph, education completion disparities between regions are lesser than disparities between financial quintiles. Therefore, an increase in the agricultural GDP decreases the region disparities, but not as much as it decreases the financial disparities.

Evaluation

Findings

This investigation has found a clear inverse relationship between agriculture GDP (%) and education completion disparities (financial, gender and region).

Important Points

- The primary education level has the highest educational completion rates and least disparities. The effect of agriculture GDP (%) on education disparities at the primary level is negligible.
- The upper secondary education level has the least education completion rates and highest disparities. The effect of agriculture GDP(%) on education disparities is the most at the upper secondary education level.
- The most education completion disparities are between financial quintiles. Agriculture GDP (%) has the most effect on financial disparities.

Shortcomings and Future Action

- In the education dataset, the values for most years are empty. Therefore, mean values were calculated for all years. A future investigation should use a dataset which contains yearly data, to carry out a time-series analysis which could provide more interesting insights.
- The datasets contain countries. This analysis calculated mean values for all the years of a specific country. This is probably an oversimplification, because countries may contain numerous regions with a very diverse set of disparities. For future direction, a dataset should be used which contains separate values for each year, and sub-region.
- This analysis used agriculture, forestry and fishing GDP(%) as a measure of the agriculture GDP(%). Although this provides the right direction, it does not exactly equate to agriculture GDP(%), possibly introducing inaccuracies. In future investigations, the exact agriculture GDP (%) should be used.
- This analysis used only one data source, the World Bank. It is possible for this data source to contain errors. Hence, for a future investigation, the data should be validated against another source.

Ethical Considerations

For this project, the following points ensure adherence to ethics during the data acquistion phase and the data processing pipeline:

- The agriculture dataset was taken from the World Bank database World Development Indicators (WDI), and the education dataset was taken from the World Bank database Education Statistics – All Indicators. Both these databases are licensed by the Creative Commons Attribution 4.0 (CC-BY 4.0). Both of these databases are available for public use. The data can be accessed easily through the notebook.
- Steps in the data processing pipeline have been clearly highlighted through a combination of markdown cells and comments. New dataframes are created for each main task in the pipeline (e.g. cleaning/structuring) as well as sub-tasks (e.g. within the task of cleaning, there are seperate dataframes for subtasks like removing outliers, coercing data

to numeric values etc.). This maintenance of seperate dataframes adheres to ethics as it provides a traceable and transparent account of how the data was modified at each stage. Additionally, this preserves the integrity of the state of data at various steps in the pipeline.

- The datasets contain information from different countries, which is categorized according to groups of people (e.g. financial quintiles, education level, region, gender). The datasets do not contain any information or PII specific to certain individuals, hence anonymization was not included as a step in the data processing pipeline. Checking the datasets for information specific to individuals adheres to ethical principles. If a dataset did contain PII, it would have to be anonymized accordingly.

When using this analysis, the following ethical points must be considered:

- This analysis used agriculture, forestry and fishing GDP(%) as a measure of the agriculture GDP(%). Although this provides the right direction, it does not exactly equate to agriculture GDP(%), possibly introducing inaccuracies. In future investigations, the exact agriculture GDP (%) should be used.
- This analysis focused on a set of specific disparities (financial, region, gender). The relationship between agricultural GDP and educational disparities was derived based on these disparity categories. Perhaps this relationship is quite different for other kinds of disparities (culture, religion etc.). Therefore, this analysis may result in bias and a misjudgment of the disparities.
- The data for each country is collected under different conditions, which may introduce bias.

Conclusion

The aim of this project has successfully been met. This exploratory data analysis investigated the relationship between agriculture GDP percentage and education disparities (financial, gender and region) across education levels (primary, lower secondary and upper secondary) in countries with a major employment percentage in agriculture from 2005 till 2023. This analysis discovered clear relationships between agriculture GDP and education disparities. This discovery could add value to the ongoing global discussion about reducing education disparities. If the future action described above is taken for this investigation, it can provide more valuable insights and be utilized in government policy making.

References

- Hanushek, E. A. & Wößmann, L., 2007. Education Quality and Economic Growth. World Bank.
- Pokropek, A., Borgonovi, F. & Jakubowski, M., 2015. Socio-economic disparities in academic achievement: A comparative analysis of mechanisms and pathways. Learning and Individual Differences.
- SHORT, C., 2002. Better Livelihoods for Poor People: The Role of Agriculture. Department for International Development.
- World Bank, n.d. Data Bank Education Statistics - All Indicators. [Online] Available at: https://databank.worldbank.org/source/education-statistics-%5E-all-indicators
- World Bank, n.d. Data Bank World Development Indicators. [Online] Available at: https://databank.worldbank.org/source/world-development-indicators